

VENDOR PERFORMANCE ANALYTICS

PYTHON | SQL | POWER BI | STATISTICAL TESTING

1. BUSINESS PROBLEM & PROJECT SCOPE

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. The enterprise must identify financial leakages caused by inefficient pricing, high holding costs, and vendor dependencies.

Primary Objectives

- Identify brands requiring promotional adjustments.
- Map vendor contributions to total Gross Profit.
- Quantify cost-reduction via bulk purchasing.

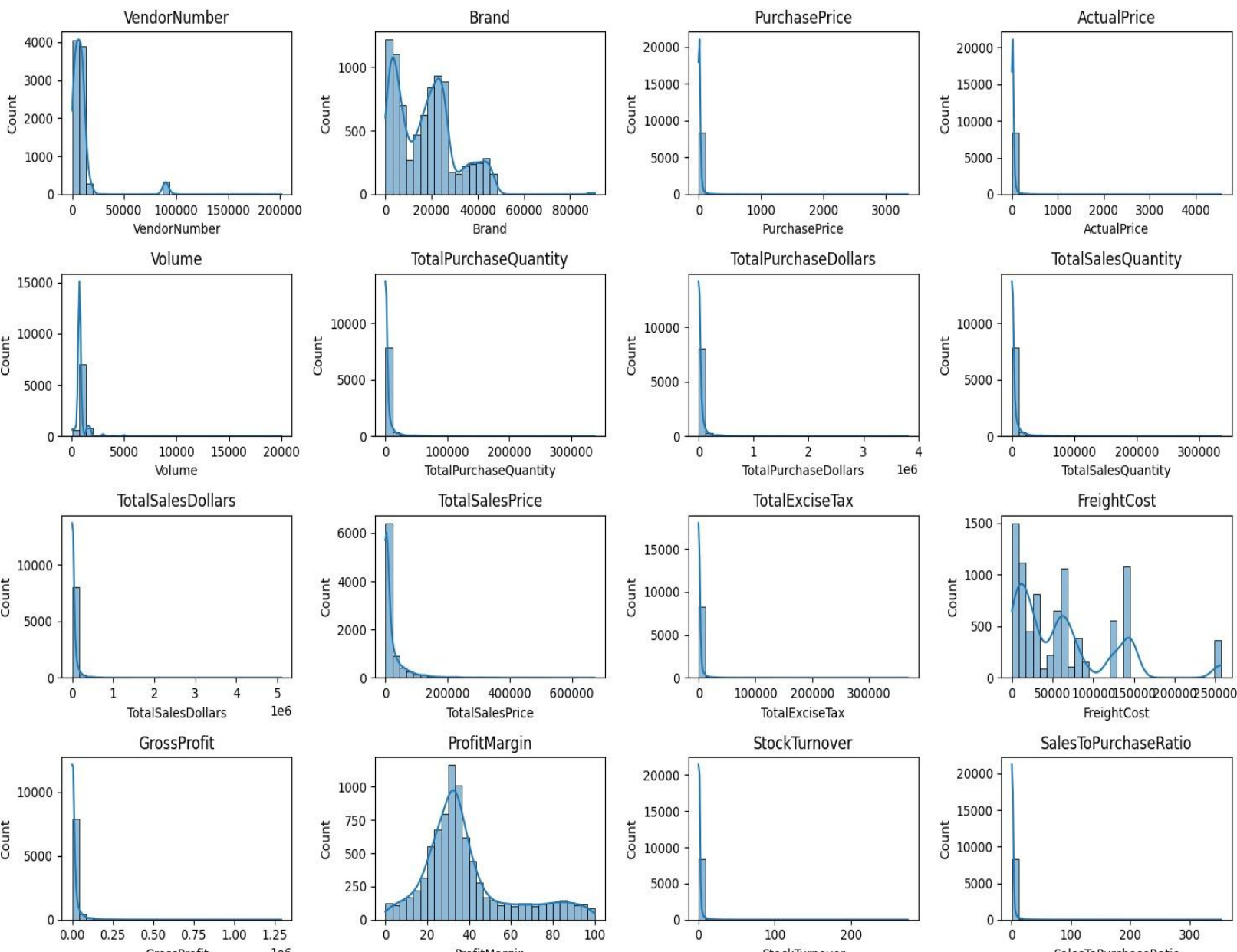
Operational Focus

- Optimize slow-moving stock turnover ratios.
- Analyze variance in profitability models.
- Validate insights using statistical T-Testing.

2. EXPLORATORY DATA ANALYSIS: SUMMARY STATISTICS

Variable	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.0000	7153.0000	9552.0000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.5000	18761.5000	25514.2500	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.8400	10.4550	19.4825	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.9900	15.9900	28.9900	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.0000	750.0000	750.0000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.0000	262.0000	1975.7500	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.4575	3655.4650	20738.2450	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.0000	261.0000	1929.2500	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.2200	5298.0450	28396.9150	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.7100	2857.8000	16059.5625	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.8000	46.5700	418.6500	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.8700	50293.6200	79528.9900	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.9200	1399.6400	8660.2000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.3245	30.4054	39.9561	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.8072	0.9815	1.0393	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.1537	1.4368	1.6654	3.529286e+02

3. DATA DISTRIBUTIONS & INSIGHTS



Primary Observations

Negative Profitability: Min Gross Profit (-\$52k) indicates products sold significantly below acquisition costs.

Dead Stock: Zero sales quantity for certain purchased SKU confirms capital lock-up.

Premium Outliers: Max Purchase Price (\$5.6k) vs. Mean (\$24) highlights a niche high-end segment.

4. DATA FILTERING & CLEANING LOGIC

To ensure the reliability of the insights, inconsistent data points were removed based on the following logic:

Step 01

Gross Profit > 0:

Excluded transactions resulting in losses to focus on growth drivers.

Step 02

Profit Margin > 0:

Removed zero/negative revenue instances to stabilize margin trends.

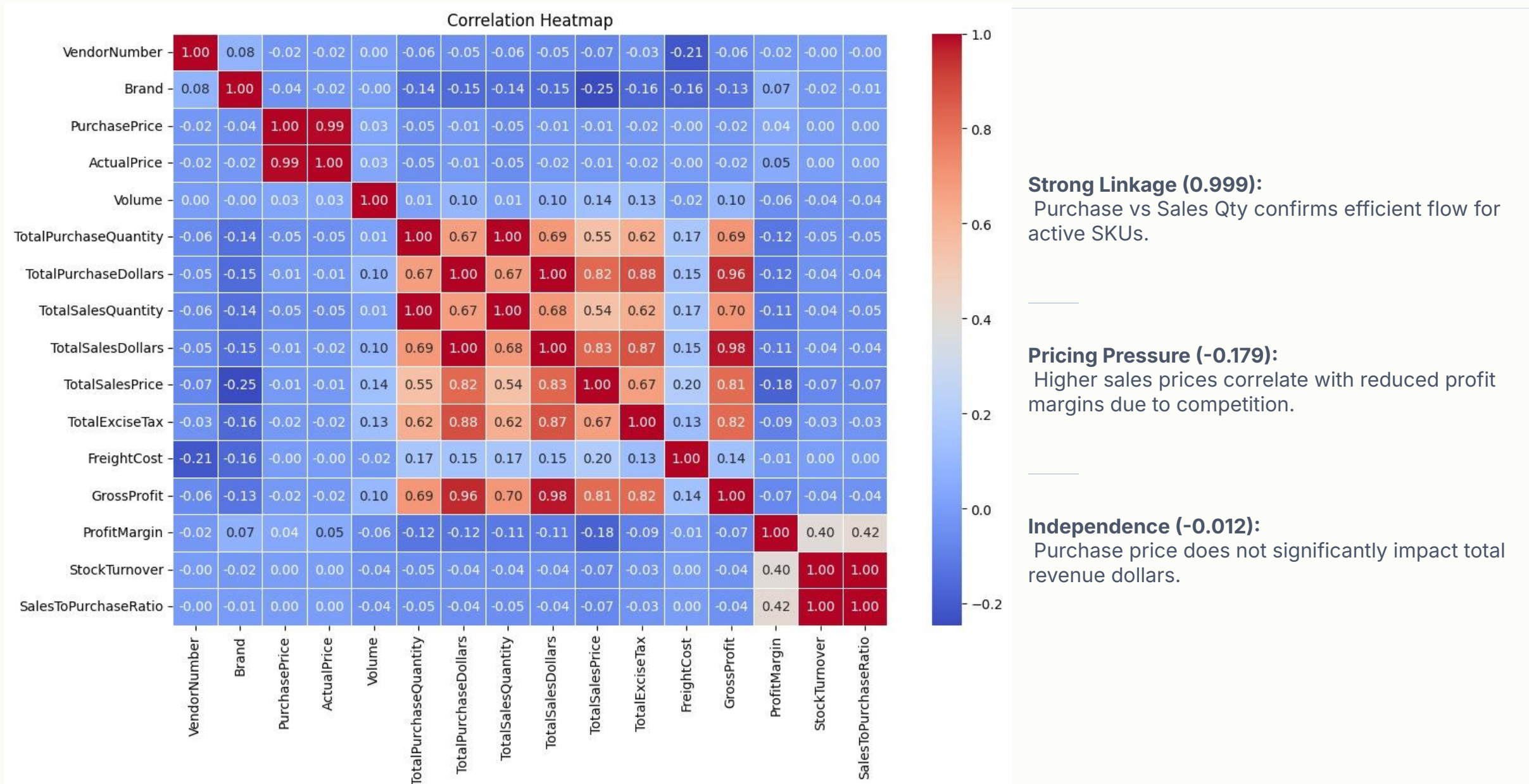
Step 03

Sales Quantity > 0:

Eliminated inventory purchased but never reached consumers.

Note: Freight costs exhibited extreme variance (\$0.09 to \$257k), suggesting logistics inefficiencies or massive bulk shipment volatility.

5. STATISTICAL CORRELATION INSIGHTS

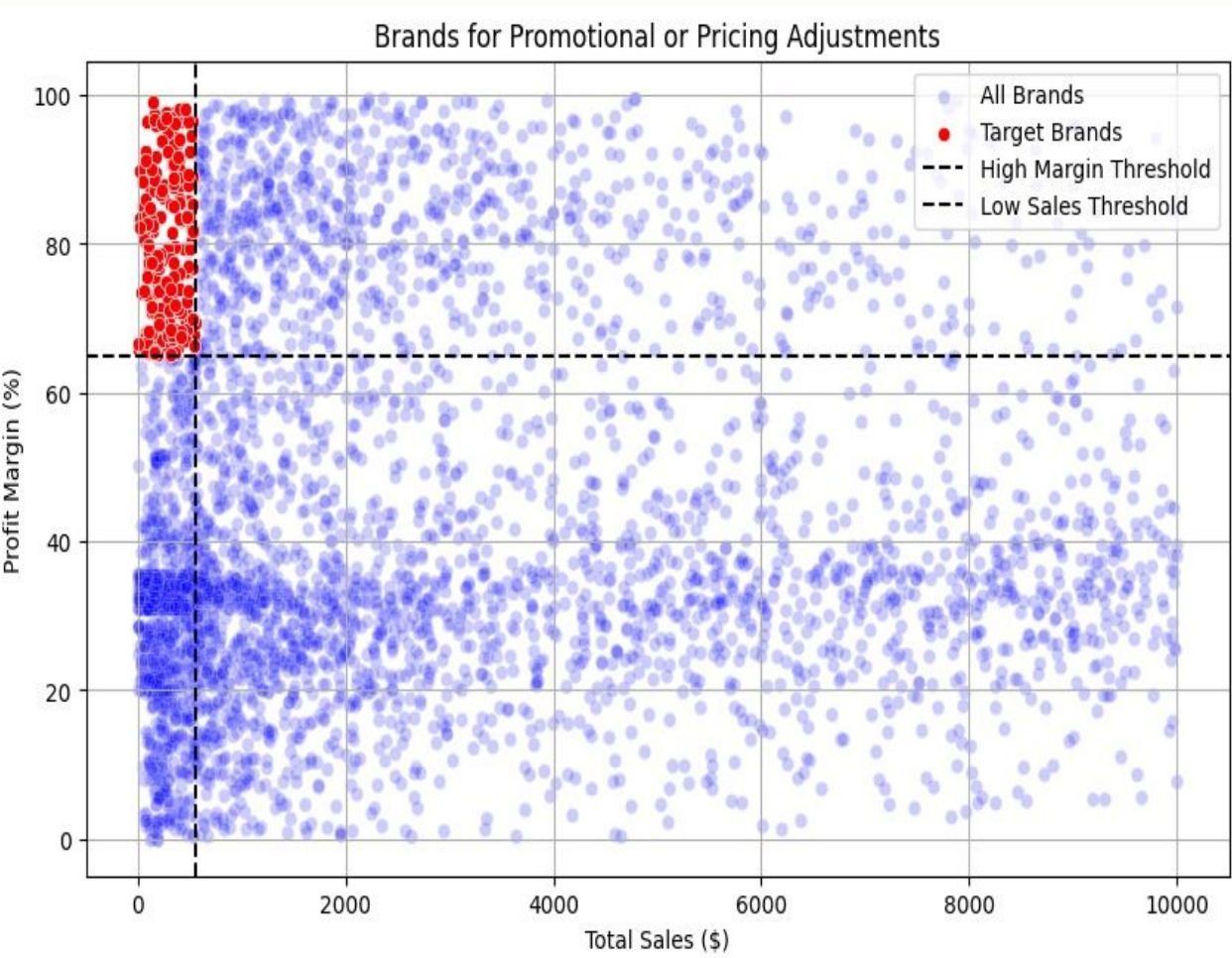


SECTION 02

RESEARCH QUESTIONS & KEY FINDINGS

Addressing the core business objectives through data-driven analysis, statistical testing, and actionable operational insights.

6. BRANDS FOR PROMOTIONAL OR PRICING ADJUSTMENTS



Identified: 198 Growth Targets

Brands with Low Sales but High Margins. Opportunity to increase volume via promotion.

Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748

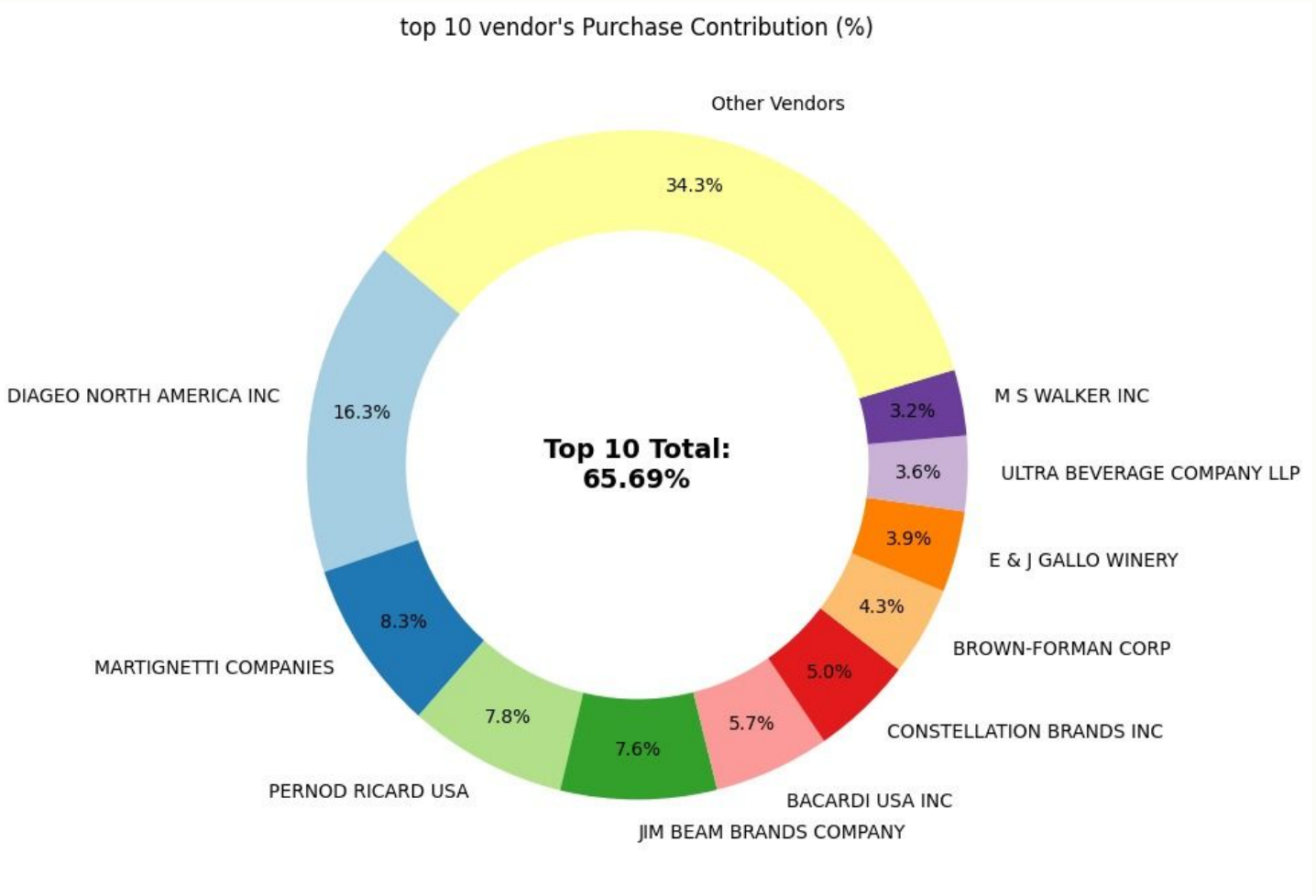
7. TOP VENDOR PURCHASE CONTRIBUTION

TOP 10 TOTAL SHARE

65.69%

Supply chain is highly concentrated. Dependence on Top 10 partners creates systemic risk. Remaining vendors contribute 34.31%.

Vendor Name	Contrib (%)
DIAGEO NORTH AMERICA INC	16.3%
MS WALKER INC	8.3%
CONSTELLATION BRANDS INC	7.8%



8. IMPACT OF BULK PURCHASING ON COST

72%

COST REDUCTION

Achieved in Large Orders vs Small Orders

Order Size	Unit Purchase Price
Small	\$39.06
Medium	\$15.49
Large	\$10.78

9. VENDORS WITH LOW INVENTORY TURNOVER

UNSOLD INVENTORY CAPITAL

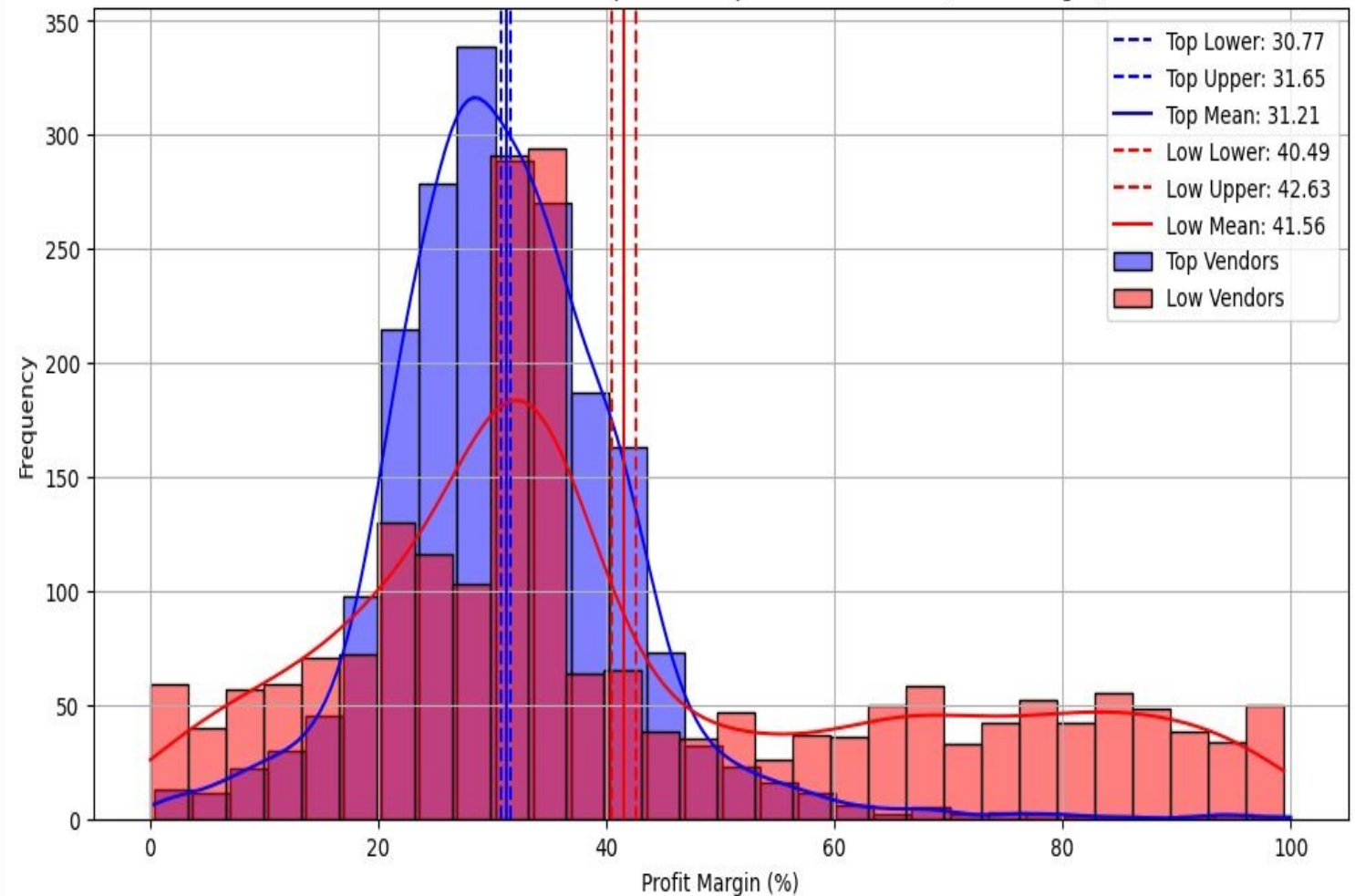
\$2.71 Million

- ❖ Slow-moving inventory increase storage costs, reduces cash flow efficiency and affects overall profitability.
- ❖ Identifying vendors with low inventory turnover enables better stock management , minimizing financial strain.

VendorName	StockTurnover	VendorName	UnsoldInventoryValue
ALISA CARR BEVERAGES	0.615385	DIAGEO NORTH AMERICA INC	722.21K
HIGHLAND WINE MERCHANTS LLC	0.708333	JIM BEAM BRANDS COMPANY	554.67K
PARK STREET IMPORTS LLC	0.751306	PERNOD RICARD USA	470.63K
Circa Wines	0.755676	WILLIAM GRANT & SONS INC	401.96K
Dunn Wine Brokers	0.766022	E & J GALLO WINERY	228.28K
CENTEUR IMPORTS LLC	0.773953	SAZERAC CO INC	198.44K
SMOKY QUARTZ DISTILLERY LLC	0.783835	BROWN-FORMAN CORP	177.73K
TAMWORTH DISTILLING	0.797078	CONSTELLATION BRANDS INC	133.62K
		MOET HENNESSY USA INC	126.48K

10. PROFIT MARGIN COMPARISON

Confidence Interval Comparison: Top vs. Low Vendors (Profit Margin)



Top Vendors Mean: 31.17%
CI: (30.74% - 31.61%)

Low Vendors Mean: 41.55%
CI: (40.48% - 42.62%)

Low-performing vendors maintain significantly higher margins but lack sales volume push.

11. STATISTICAL VALIDATION

T-Test for Profit Margin Delta

H_0 : No difference in margins between top/low vendors.
 H_1 : Significant difference exists.

RESULT: REJECT NULL

Statistically confirmed: Groups operate under distinct profitability models.

12. STRATEGIC RECOMMENDATIONS

- **Pricing:** Promotional push for 198 high-margin niche brands.
 - **Diversification:** Reduce 65% dependency on top vendors.
 - **Capital:** Liquidate \$2.71M unsold stock at slow vendors.
 - **Bulk:** Consolidate smaller orders to hit the 72% cost savings mark.
 - **Growth:** Focus marketing on high-margin/low-sales brands.
 - **Efficiency:** Monitor freight cost volatility in logistics.
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